

GLOBAL SUPPLY CHAIN & LOGISTICS PERFORMANCE

Submitted By

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1. Introduction

Global Supply Chain & Logistics Performance focuses on monitoring, analyzing, and improving the movement of goods from suppliers to customers. Efficient supply chain management helps organizations reduce costs, improve delivery performance, optimize inventory levels, and enhance customer satisfaction.

2. Problem Statement

Organizations face challenges such as shipment delays, inventory shortages, increasing transportation costs, and inefficient logistics operations. Analyzing supply chain data can help businesses identify operational issues and improve overall performance.

3. Objectives

- Understand global supply chain operations.
- Analyze shipment and transportation performance.
- Identify delivery delays.
- Evaluate inventory and warehouse efficiency.
- Develop analytical dashboards and reports.

4. Scope of the Project

- Manufacturing Industries
- E-Commerce Companies
- Retail Supply Chains
- Logistics Service Providers
- Warehousing Operations

5. Technologies Used

Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Google Drive

6. System Architecture

Logistics Dataset → Data Collection → Data Cleaning → Data Analysis → Performance Evaluation → Visualization Dashboard → Recommendations

7. Methodology

Data Collection, Data Preprocessing, Exploratory Data Analysis, Performance Evaluation, Visualization, and Recommendation.

8. Dataset Features

Shipment ID, Order ID, Product Category, Origin Location, Destination Location, Shipping Mode, Delivery Status, Transportation Cost, Inventory Level, Warehouse Capacity, Delivery Time.

9. Analytics Techniques

Descriptive Analytics, Data Visualization, Trend Analysis, Performance Monitoring, KPI Evaluation.

10. Expected Output

Shipment performance reports, delay analysis reports, inventory monitoring dashboards, transportation cost analysis, and optimization recommendations.

11. Benefits

Reduces transportation costs, improves inventory management, minimizes shipment delays, and enhances customer satisfaction.

12. Future Enhancements

Real-time logistics monitoring, AI-based demand forecasting, predictive inventory management, and cloud-based analytics dashboards.

13. Conclusion

The project helps organizations analyze logistics operations and improve supply chain efficiency using data analytics and visualization techniques.

14. References

Python Documentation, Pandas Documentation, Logistics Analytics Research Papers, Supply Chain Management Case Studies, Google Colab Documentation.